

- 10 (a) Radioactive decay is a spontaneous process.

State the meaning, in this context, of the term spontaneous.

.....
 [1]

- (b) Two radioactive isotopes X and Y each decay to form a stable isotope.
 A sample initially contains only atoms of isotope X. At this time, its activity is $4A$.
 Another sample initially contains only atoms of Y. At this time, its activity is A .

Fig. 10.1 shows the variation of the activity of each sample with time t between $t = 0$ and $t = 6T$.

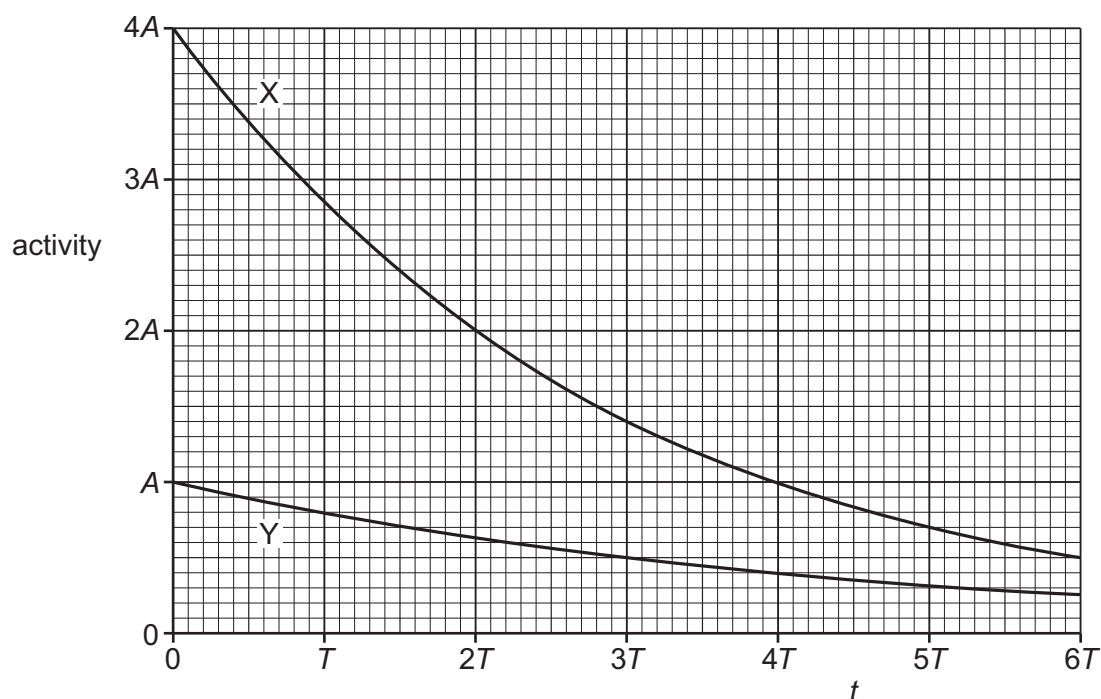


Fig. 10.1

- (i) Complete Table 10.1 to give expressions, in terms of either or both of A and T , for the quantities indicated for each of the samples.

Table 10.1

sample	half-life	decay constant	initial activity	initial number of nuclei
X			$4A$	
Y			A	

[3]

